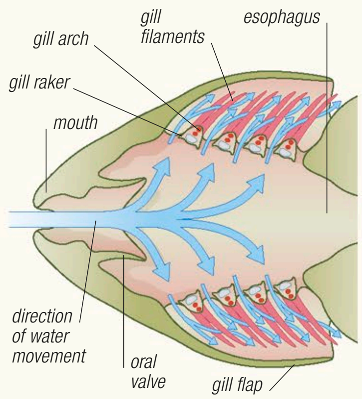
GILLS

A fish's gills are located in internal chambers on either side of the body, just behind the mouth. In most species, each gill consists of a framework of bone or cartilage, supporting tissues that contain densely packed capillaries. As water enters the gill chambers, it is first sieved clean by structures called gill rakers. Behind the rakers are supports called gill arches, which provide attachment points for gill filaments. To increase the efficiency of respiration, the filaments have folds on their surface, known as lamellae. Gases are exchanged across the surface of the lamellae, before being distributed to the rest of the fish's body. In some fishes, the area of the lamellae is 10 times greater than that of the body's outer surface.



GAS EXCHANGE
Water enters a fish's gill chamber through the mouth (or in some, the spiracles) and leaves through the gill openings. As it passes through the gills, dissolved oxygen is transferred across the thin outer membrane of the capillaries and enters the bloodstream. At the same time, carbon dioxide is expelled from the capillaries.

BREATHING OUT OF WATER
By keeping their gill chambers moist, mudskippers can survive out of water for long periods. They live in coastal swamps, and at low tide can be found using their well-developed pectoral fins to move around on exposed mudflats.

have a skeleton of cartilage hardened by calcium carbonate, as well as a fully developed vertebral column. Most fishes have a skeleton made of bone. In addition to a bony skull, vertebral column, and skeleton, they have bony supports in the fins.

Although some fishes have smooth skin, most are protected by scales, plates of bone, or spines. Scales, which are the most common covering, provide protection and promote the efficient movement of water over the body, while still allowing the fish to move freely (see panel, opposite). Glands in the skin secrete mucus, which protects the fish from bacteria and, in some species, helps to reduce drag.

Almost all fishes have fins. There are 2 main types: median (or unpaired) and paired fins. Median fins are found singly, on the dorsal or ventral midline of the body. They include the dorsal, anal, and tail (caudal) fins. The paired fins—the pelvic and pectoral fins—are arranged in twos, one on each side of the body. Fins are used mainly for locomotion (see p.470) but in ray-finned fishes they also have other uses: colors and patterns on their surface can be used as signals to warn predators, attract mates, defend territories, or lure prey; the fins of some fishes have spines, which are sometimes poisonous, to protect them from predators.

All fishes have gills, which they use to take oxygen into their bloodstream (see panel, above). Some fishes also have other ways of taking in oxygen. Lungfishes, for example, breathe air using primitive, lunglike organs. Other fishes can absorb oxygen and release carbon dioxide through their skin.

Senses

Fishes collect information about their environment using sensory systems commonly found in other types of animals—these include vision, hearing, touch, taste, and smell. However, fishes also have some unique sense organs.

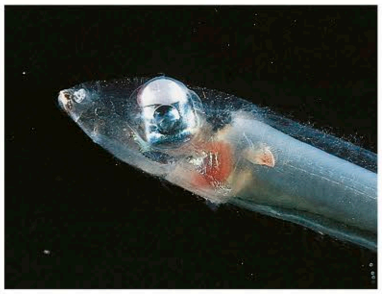
Although most fishes have eyes, their ability to perceive light, color, shape, and distance varies greatly between species. This variation can often be related to habitat. Fishes that live in clear water tend to have good vision, while many fishes that live in dark conditions (such as muddy water, caves, or the deep sea) either have poor vision or have lost their eyes altogether. However, some deep-sea fishes have large eyes so that they can collect as much light as possible.

Water is an effective medium for transmitting sound, and most fishes are able to respond to sound waves. Sound vibrations are usually transmitted through the bones and tissues of the head to the inner ear, although in some species they may be amplified by the swim bladder.

In aquatic animals, the distinction between taste and smell is blurred because the same sense organs often respond to chemicals that are dissolved in water and contained in food. Fishes smell using sensory pits (called nares), which are lined with tissues that contain olfactory receptor cells. Most fishes have a good sense of smell, and some can detect chemicals in minute concentration. Taste receptors are usually found in or near the mouth, although some fishes have them on their fins, skin, or whiskerlike barbels, allowing them to taste food on or near the bottom.

Almost all fishes have what is called a lateral-line system, which they use to detect vibrations (including those made by their predators and prey) and changes in water pressure and currents. These changes are detected by organs known as neuromasts, which are usually housed in canals, in the bones of the head and in a lateral line canal that runs along the side of the body beneath the scales and is connected to the exterior via pores that run through the scales.

Many fishes are also able to detect electrical waves and impulses. In cartilaginous fishes, such as sharks, signals are received by structures called the ampullae of Lorenzini. These specialized nerve cells, located in small pores at the surface of the skin, contain a conductive gel and can detect the weak electrical currents produced by other animals. Some ray-finned fish can also detect electrical signals. Certain species have organs that produce electrical fields, which they use to detect objects or other fish in low visibility, and to communicate.



TUBULAR EYES

LARGE EYES

EYES
Fishes that live in habitats with little light often have unusual eyes. Spookfishes (top) are deep-water fishes with tubular eyes that point upward. The pinecone soldierfish (above) lives in shallow water but avoids direct sunlight. It feeds at night, using its large eyes to search for food.

TASTE
Catfishes are named after the whiskerlike barbels found in many species. These are covered by taste buds and are used to find food. This is especially useful for species such as the squeaker (shown here), which lives in dark waters at the bottom of deep lakes.